

LeetCode 1473: Paint House III

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1. Problem Understanding & Definitions

Key terms and array representations given in the problem:

`houses[i]` → colour of house i and 0 if it's not yet painted , and can only paint if not already painted

`cost[i][j]` → cost of painting i^{th} house with color $(j+1)$

Neighborhood: a neighborhood is blocks of houses having same color.

Eg: **1** | **2 2** | **3 3** | **2** | **1 1**

→ **5 neighborhood**

Objective: we need to repaint to create *target* number of neighborhood.

Output: *min cost*

Example Execution:

`houses = [0, 0, 0, 0, 0]`

`cost = [[1, 10], [10, 1], [10, 1], [1, 10], [5, 1]]`

Optimal Costs Chosen: $1 + 1 + 1 + 1 + 5 = 9$

2. Dynamic Programming State & Transitions

Colour of the houses matter too , cant simply draw partitions

Therefore, we define the DP state as:

```
dp[i][color][groups]
```

Transitions:

```
dp[i-1][prev_colour]  
[groups]
```

If color same as prev color

'OR'

```
dp[i-1][prev_colour]  
[groups - 1]
```

If color diff from prev color

NOTE:

If $\text{houses}[i-1] \neq 0 \rightarrow$ forced to use it's colour.

Otherwise try all m colours and add cost.


```

        if(prevc==c){

            if(dp[i-1][prevc][groups]==INT_MAX)
continue;

            dp[i][c][groups]=min(dp[i][c]
[groups],dp[i-1][prevc][groups]+pcost);
        }

        else{

            if(dp[i-1][prevc][groups-1]==INT_MAX)
continue;

            dp[i][c][groups]=min(dp[i][c]
[groups],dp[i-1][prevc][groups-1]+pcost);
        }
    }
}

int ans=INT_MAX;
//akhri ghar kaise bhi paint karo
for(int c=1;c<=m;c++){
    ans=min(ans,dp[n][c][target]);
}

if(ans!=INT_MAX)
return ans;

else
return -1;
}
};

```